SKIN & FASCIA: COMPREHENSIVE OVERVIEW

I. Skin

Structure

- **Epidermis:** Keratinized stratified squamous epithelium devoid of blood vessels. Stratified squamous consists of squamous (flattened) epithelial cells arranged in layers upon a basal membrane.
- **Dermis:** Connective tissue containing blood vessels, lymph vessels, sensory nerve endings, smooth muscle, hair follicles, sweat and sebaceous glands. In its deep part, collagen bundles are arranged in parallel rows.
- **Hypoderm (Superficial Fascia/Cutaneous Tissue):** Deep to the dermis.

Functions

- **Protection:** Abrasion, invasion, water loss, UV protection.
- **Vitamin D synthesis:** Epidermal keratinocytes when exposed to UV light help maintain skeleton health by increasing Ca2+ absorption.
- **Sensation:** Receptors for heat, cold, touch, pressure, vibration, pain.
- **Thermoregulation:** Thermo receptors and sweat glands, hypothalamus controls cutaneous arteries and sweat glands to retain or dissipate heat.
- **Psychological and social functions:** Appearance, social acceptance, facial expression, nonverbal communication.

II. Fascia

Fascia is a tough, continuous connective tissue that covers and interpenetrates every muscle, bone, nerve, artery, vein, organ, and cell in the body.

1. Superficial Fascia

Feature	Description
Composition	Loose, mixture of adipose and loose areolar tissues.
Function	Unites skin to underlying structures, facilitates movement, passage for cutaneous vessels/ nerves, protects against heat loss.
Density Variations	Dense in scalp, palm, sole. Thin in eyelids, auricle, scrotum, penis, clitoris (devoid of fat).
Contents	Fat, vascular structures (capillaries, lymphatics), nervous tissues (Pacinian corpuscles).
Fat Distribution	Very less: Eyelids, penis. More: Breast, abdomen, gluteal region.

2. Deep Fascia

Type/Feature	Description
General Characteristics	More dense, compact, and regularly arranged collagenous bundles than superficial. Present as membranes.
Investing Fascia	Covers muscle surfaces. In neck, forms well-defined layers; in limbs, forms a definite sheath.
Intermuscular Septa	Lie between muscles, dividing limbs into compartments.
Retinacula	Prevent bowstringing.
Other Structures	Fibrous sheaths (Carotid/Axillary), fibrous capsules (Parotid), ligaments.
Absence Areas	Face, Breast, Penis, Anterior abdominal wall.

3. Subserous Fascia

Feature	Description
Composition	Loose areolar tissue.
Function	Covers internal visceral organs. Provides lubricating fluid (via circulatory channels) for movement and nutrition.

III. Molecular & Biomechanical Characteristics

- **Components:** Collagen (strength/anti-overextension), Elastin (elasticity), Ground substance (polysaccharide gel, minimizes friction).
- **Biomechanical Characteristics:** Elasticity (retain shape), Creep (relaxation under constant load), Hysteresis (capacity to lose energy under stress).
- **Clinical Considerations:** Trauma/inflammation can cause fascial knitting (excessive pressure). Injury causes inflammatory processes; in tight deep fascia compartments, this leads to *Compartment Syndrome*.

Biochemical and immunological changes occur within the ground substance and have general systemic effects that seem quiet far removed from the injury to the soft tissues .

Biomechanical Characteristics

- **Elasticity:** Ability to retain shape and respond to deformation.
- **Creep:** Relaxation under sustained load, reducing resistance to subsequent loading.
- **Hysteresis:** Energy loss during stress cycles; a principle used in myofascial release.

Clinical Significance

Injury, poor posture, or inflammation can cause fascia to become restricted, increasing pressure on internal structures. The inflammatory process in deep fascial compartments may lead to **Compartment Syndrome** due to the inability of the dense tissue to expand.

Function of Fascia

- It provides support for vessels and nerves throughout the body
- It enables adjacent tissues to move upon each other while providing stability and contour.
- It provides lubricating fluid between structures for movement and nutrition
- The bacinian corpiscles in the superficial fascia provide afferent information to spinal cord and brain
- Within the ground substance of the fascia are many of the substances that contribute to immune mechanisms within the body